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Unit 4 : Session 3 : SLO 1 | Custom Higher-Order Functions

I. Fill in the Blanks

1. lambda a lambda function in Python is a short anonymous function.
2. function a higher-order function can take another function as an argument (or return one).

II. Multiple Choice

1. `list(filter(lambda x: x % 2 == 0, [1, 2, 3, 4]))` [2, 4] filter keeps only the even numbers.

III. Example Custom Higher-Order Function

- `def apply_twice(func, value): return func(func(value))`
- `apply_twice(lambda x: x * 3, 2)` # 18
- Higher-order functions treat functions as first-class values: they can be passed, returned and stored like any other object.